

Python uv Cheat Sheet: Virtual Environments & Package Management

What is uv?

uv is a fast, modern Python package and virtual environment manager. It replaces common workflows involving pip and venv with significantly faster dependency resolution.

Installing uv

```
pip install uv
```

macOS/Linux alternative:

```
curl -Ls https://astral.sh/uv/install.sh | sh
```

Creating Virtual Environments

Create environment in current project:

```
uv venv .venv
```

Activating Environments

Windows:

```
.venv\Scripts\activate
```

macOS/Linux:

```
source .venv/bin/activate
```

Installing Packages

```
uv pip install requests
```

```
uv pip install -r requirements.txt
```

Managing Dependencies

Freeze dependencies:

```
uv pip freeze > requirements.txt
```

Upgrade packages:

```
uv pip install --upgrade package_name
```

Remove packages:

```
uv pip uninstall package_name
```

Best Practices

- Use one environment per project
- Commit requirements.txt to version control
- Use uv for faster CI and local installs
- Avoid mixing pip and uv in the same workflow